AN INTELLIGENT ASSISTANT FOR FERTILITY PREDICTION, WELLNESS, AND SUPPORT

Project ID – 25-26J-246

Project Proposal Report

Bachelor of Science (Hons) Degree in Information Technology Specializing in

Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology (SLIIT)

Sri Lanka

August - 2025

AN INTELLIGENT ASSISTANT FOR FERTILITY PREDICTION, WELLNESS, AND SUPPORT BEHAVIORAL & PSYCHOLOGICAL SUPPORT MODULE

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DECLARATION

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I hereby declare that the work presented in this report of proposal is my own and has not been submitted elsewhere. All sources of information have been duly acknowledged. This report is submitted in partial fulfillment of the requirements of the above degree program, and it reflects the research and development plan for my individual component of the project.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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Signature of the Supervisor Date

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..... 31/08/2025

Signature of the Co-Supervisor Date

Mrs. Vishaliney Pirathap

ABSTRACT

IVF procedures place powerful psychological and behavioral burdens on patients, such as high levels of stress, disturbed sleep, fluctuating mood, and low levels of activity, which may have a negative impact on the success of the treatment. This proposal describes the Behavioral and Psychological Support Module, which is an essential part of the AI-Powered IVF Lifestyle Support System, and it will help keep track of stress, sleep quality, activity, and mood and provide tailored reminders like meditation guides, journaling prompts, calming music suggestions, and yoga suggestions. By taking advantage of machine learning to analyze real-time and integrate wearable devices, the module will help improve emotional well-being and increase the success rates of IVF. Based on eleven available studies on maternal mental health, stress detection, and infertility psychology, this module fills a very important gap in comprehensive IVF care. This process includes sensors, AI-driven personalization, and user feedback loops as the methodologies. The anticipated results are lower stress levels and increased compliance with lifestyle changes, and the scope is restricted to psychological assistance under IVF conditions. Work plan and references are detailed.

Keywords

IVF, Psychological Support, Stress Management, Sleep Quality, Mood Monitoring

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LIST OF ABBREVIATIONS

IVF – In Vitro Fertilization

AI – Artificial Intelligence

ML – Machine Learning

IoT – Internet of Things

ANFIS – Adaptive Neuro-Fuzzy Inference System

ART – Assisted Reproductive Technology

mHealth – Mobile Health

NLP – Natural Language Processing

UX – User Experience

1. INTRODUCTION

IVF procedures frequently create great burdens on the patient emotionally and daily, as high-stress levels, sleeplessness, and mood fluctuations lead to great challenges. Mental well-being is one of the aspects that can have significant impact on the effectiveness of the treatment. The Behavioral and Psychological Support Module is an essential component of the AI-Powered IVF Lifestyle Support System that focuses on these issues with wearable technology and machine learning to track a real-time state of stress, sleep quality, activity levels, and mood. The module will provide individualized care, including meditation scripts, journaling, relaxing music, and yoga recommendations, to improve emotional well-being and expand the outcomes of IVF. This module is designed with the requirements of IVF patients in mind and is therefore an essential gap in the provision of holistic care with specific support beyond the general health support offered to patients. It also takes advantage of modern technology to deliver the required interventions in time; this makes the patient deal with their emotional and physical status in their treatment process. One of the systems gathers information via sensors and user inputs, process it using intelligent algorithms and provide personalized advice to enhance resiliency and the ability to maintain healthy habits. The introduction sheds light on the novelty of the module, which aims at helping IVF patients through the incorporation of mental health care into their care. Later paragraphs will explore the research background outlining the objectives, the methodology, and plans of implementation. The module will help establish a more favorable environment because it considers the emotional and psychological dimension of care that conventional IVF care does not address, which may translate into a higher success rate of treatments and better patient health. This initiative is a stride towards quality care provision of fertility treatment beneficiaries.

1.1 BACKGORUND AND LITERATURE SURVEY

1.1.1 Background

Over the last years, mental health support tools have grown. Examples include mobile applications and smart devices that can monitor stress, sleep and mood in real-time. These devices detect the issues at an initial stage due to the assistance of mere sensors. Studies have shown that the stressful levels of IVF could harm the development of the baby and predispose the infant to develop sadness [1], [8]. The research on locating stress and sadness shows it as a race: the more troubles are the more the check tools should be improved[4]. The old techniques were searching at symptoms that could be easily spotted, like sleep deprivation or insufficient activity, and new complications are harder to figure out. According to the latest studies, such strategies should check body cues or mood patterns which are not standard and should be checked [5]. In one study, a smart system to find the location of stress was applied without physically touching the body. It yielded good outcomes by altering heart and moving data [3]. It means that we can develop an IVF-specific check tool with the help of real patient data to detect suspicious patterns. In the consciousness component, applications can be one-on-one like relaxation and yoga classes. We have used smart assistants to guide mothers through tough times [11]. Finding mind issues is hard. Certain check tools exist, including ones used to test the use of words to ascertain low mood[7]. Research has however shown that these tools are not necessarily always accurate, especially where issues are conspicuous with the day-to-day life[6]. In our example, a patient can be undergoing real stress yet with simplistic ways of coping. The need to be rigid in our reasoning must be a question of necessity, and successful, it may not hurt. We are concerned with applied IVF issues, including an enduring phobia regarding failed attempts[8].

1.1.2 Applications in IVF Context

At this moment, IVF aids applications do not verify completely mind well-being AI. But like regions indicate the need. In pregnancy care, sadness and stress are increasingly being discovered in research. One 2022 study examined mobile interventions on mother sadness in pregnancy. It emphasized that body/word mixing can be helpful [1]. This is suitable in our module because we have sensor data and mood notes. As an illustration, when sleeping feels good and mood is poor, that disparity may indicate a latent problem. Unwonted pairings, such as active but angry, may be an indicator. Further, studies on mind effects in fertility are that auto checks ought to interconnect human assistance in essential instances [3]. In the case of our project, we will auto check but have levels above which hard cases will visit a doctor to have a review and not necessarily cease.

1.1.3 Monitoring Techniques Overview

We consider some methods of research to test mind problems: Body Checks: Ancient methods test heart or sleep data. New AI models train models on problem vs real. One is using sensors to locate stress without tools [3]. The other uses the group checks to determine the impact of age and partner help on the mind impact [1]. Apps such as calm tools, which are available on the store, boast of being highly right to find stress as of 2024 [11]. They probably apply large volumes of data and narrow in on models to identify problems. In our module, it is possible to get outside assistance, which may be expensive or personal. Simple check we will do ourselves, perhaps training on mix of real IVF data and test cases. Marks and Data: There are certain signs that are added hidden by some tools, and bad users may delete them. We cannot count on data alone. Mood Checks: The steadiness of mood is looked upon in many ways. Normal emotions are not of the same pattern as constant low mood[11] . As an illustration, even low energy may manifest in sad

times. Other checks apply models of known problems vs normal [4] But these may bring into error normal changes which are only strong. One study claimed that it is still difficult to find mind problems [1]. Even simple cases were flagged as a new check indicated a wrong [5]. We must be attentive on our part: when we check mood, we are to take it as one indication, not as complete evidence. We will combine it with data, such as bad sleep and bad mood, that will be more powerful to indicate.

1.1.4 Existing Tools and Models

We note some tools that exist. To test stress, open models are available on sites and papers, such as a test of birth problems caused by stress [5], or a recent test of the health of the mind. To the mood, old check implements (since halted due to bad work) and fresh research, such as those that examine word changes to determine whether issue probable[9] . We can also guess the possibility, should we believe that there are some problems with users, by using models. This is possible (check steady under model) and expensive. However, in practice, under hard, we may apply our simple rule: odd pattern with very low mood may be flagged but not counted only on it. To summarize, research and instruments demonstrate: “It is possible to check stress with good rightness using new models, particularly when those are prepared by area. Mood checking is not so certain but mix ways and levels can provide some find (with know of wrong flag risk)[10].

1.2 Research Gap

The main gap in research and application is the use of mind checks and artificial intelligence in IVF lifestyle support systems. Previous studies on stress and sadness mainly focused on pregnancy or on general health, not on fertility applications. In our project, the identified gap is that no existing IVF application monitors whether patient mood or stress data is real or missing. Most systems assume the data given is true, but this can be incorrect, especially with increasing mental health problems. This shows a gap in safe methods for handling patient data against hidden issues. For example, user notes have not been screened for deeper psychological problems. Older systems mostly concentrate on basic health, rather than checking whether mood has remained low for a long time. The situation is sensitive because a bad mood may not always look serious in words, but it can still hide real pain, which is harmful. From the research point of view, the combination of body signals and mood signals is a novel approach. Although there are some reports that this kind of combined mind check is developing [1], there is still limited literature on joining sensor data with mood information for decision-making, especially in specific areas such as IVF stress. Another weakness in earlier research is the management of outputs in practical use. Many studies stop at simply identifying a number or result. In a working system, we need defined levels and actions. For example, when stress is high, support should be stopped and flagged. When stress is at a medium level, the system should send the case for doctor review. When stress is low, the system should allow the process to continue but keep monitoring. This type of practical guidance is missing in past studies, and our project aims to correct it by building clear rules.

In summary, the distance between new mind checks methods and their actual use in IVF applications is still large. This shows two important needs: adapting checks to IVF patient needs and building complete system connections (from data input to flagged results and user notes). Since there is no earlier fertility application that filters mental health issues [6], our project is an important first step in addressing this gap [6].

1.3 Research Problem

The area of research in this section describes how the system can identify automatically whether data in an IVF report of stress, sleep, activity, and mood are truly a reflection of the actual conditions in a patient or are the result of lack of or falsification of information, and how it reacts to preserve the integrity of data. Patterns of abnormal stress or abnormal sleep conditions are recognized by the system as being associated with actual health conditions, and false data are corrected by recognizing genuine records and misplaced or falsely entered records. It determines whether a mood entry reflects a genuine low mood or a transient fluctuation, based on cues such as the detection of the absence of the personal pronouns in the notes or the numerical patterns in the trained models. The system compares body information to mood information to create checks and balances, e.g. high stress must correspond to a low mood, a suspected error in reporting should be signaled as an error, not as a medical problem. The checks are also adjusted to identify actual issues with minimum false alerts, which have been refined by trial-and-error adjustments. When anomalies are found, the system provides automatic feedback to the patient, including reminding her to confirm odd data or notify a doctor to inspect it, informed by best practice across different systems. Technically, the method builds two checks, the body signals B and the mood inputs M, with each generating an authenticity score. A function F gives signal B and mood M whether the joint information is accepted or flagged, depending on minimizing practice errors. This approach fosters confidence in patient information, which will improve the process of validation by modifying and experimenting with the existing models in the IVF domain.

2. OBJECTIVES

2.1 Main Objective

design an AI-powered Behavioral and Psychological Support Module, which constantly keeps track of the emotional and behavioral health of women undergoing IVF, and recommends individually tailored interventions that minimize stress, enhance mental resilience, and promote successful treatment.

2.2 Specific Objectives

Physiological & Behavioral Data Collection

To design a mechanism for collecting real-time data such as heart rate variability, sleep quality, daily activity, and mood self-assessments through wearable devices and user inputs. This ensures the system has reliable information about the patient's physical and emotional state.

AI-Based Stress and Mood Detection

To develop machine learning models that can analyze physiological signals, and mood reports to detect patterns of stress, anxiety, and emotional fluctuations. The aim is to identify changes early and provide timely support to the patient.

Personalized Intervention Generation

To create a recommendation engine that provides tailored suggestions such as meditation exercises, journaling prompts, or calming music. These recommendations will be context-aware, meaning they respond directly to the patient's detected stress or mood level.

Multimodal Consistency & Feedback Loop

To integrate both wearable data and user-reported mood inputs for a more accurate emotional profile. The module will also include a feedback system where user responses to interventions are recorded, allowing the AI to adapt and improve future recommendations.

Integration with IVF App Workflow

To ensure seamless integration of the support module with the larger IVF assistance application. The module should work alongside other features such as diet planning and treatment tracking, offering patients a holistic and user-friendly digital health solution.

Performance, Usability & Privacy Assurance

To optimize the system for quick responses and high accuracy while maintaining a user-friendly interface. At the same time, the module must comply with healthcare privacy standards, ensuring that sensitive patient data is protected through encryption and ethical data handling practices.

3 METHODOLOGY

Developing the Behavioral and Psychological Support Module has a systematic approach to accuracy, usability and effectiveness. The initial one is to clarify the scope and requirements defining the purpose of the module, its target audience (IVF patients), and the overall goals like stress detection, emotional monitoring, and personal support. Design stage is then concerned with the system structure. This involves developing the data flow, choosing the right AI models to be used in predicting stress and designing the user interface with easy-to-understand dashboards, mood journals and recommendation displays. It is also designed in a manner that is culturally appropriate and user friendly to the Sri Lankan users. The technology selection step implies the selection of wearable devices as a source of physiological data (heart rate variability, sleep quality, stress levels), the mobile application platform (React Native), the back-end technologies (Python, Flask/Django), and the cloud services to store and deploy. The implementation phase combines elements:(1) wearable data gathering, (2) mood self-Testing, (3) AI/ML stress determination, (4) recommendation (meditation, journaling, and relaxation exercises) functions. After implementation, testing and refining of the module is done. This will involve user testing where volunteers are used, feedback, and the accuracy of the stress detection and usefulness of the recommendations. Once successfully tested, the system is rolled out and performance is monitored in the real world. New updates will be made continuously to resolve the problems, improve services, and add new technologies or intervention strategies. This systematic approach will make sure that the module is a source of credible, helpful, and easy-to-use psychological support throughout the IVF process.

4 GANTT CHART

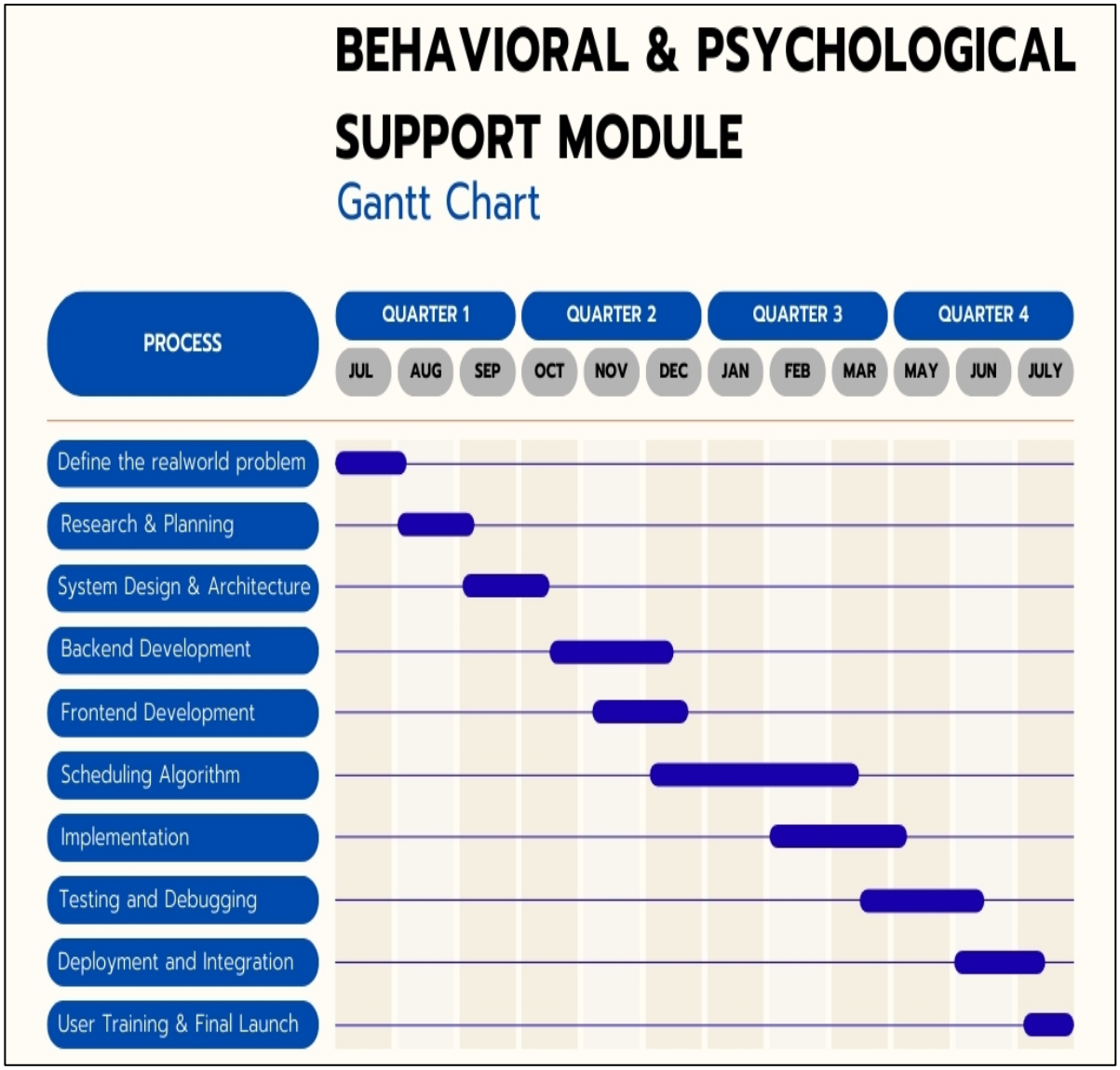

Figure 4.1 Gantt Chart

5 PROJECT REQUIREMENTS

5.1 Functional Requirements

The development of the Behavioral and Psychological Support Module follows a systematic approach to ensure accuracy, usability, and effectiveness.

The first step is to clarify the scope and requirements by defining the purpose of the module, the target audience (IVF patients), and the overall goals such as stress detection, emotional monitoring, and personal support.

The design stage focuses on the system structure, including the development of data flow and the selection of suitable AI models for predicting stress.

During design, the user interface is created with easy-to-understand dashboards, mood journals, and recommendation displays.

The design is made to be culturally appropriate and user-friendly, especially for Sri Lankan users.

The technology selection stage involves identifying wearable devices to collect physiological data such as heart rate variability, sleep quality, and stress levels.

The mobile application platform is chosen as React Native, while the back-end technologies include Python with Flask or Django

5.2 Non-Functional Requirements

Accuracy and Reliability

The checking tools should avoid raising wrong flags on normal data.

The target is to keep wrong flags under 5% when tested with normal data samples.

The module should successfully detect around 90% of obvious stress or low mood cases.

Performance

The data checks should not create long waiting times for users.

Stress, sleep, activity, and mood check together should run in less than 2 seconds on the server for one session.

If large AI models are used, they should be optimized to run faster or shifted to background processing.

The design plan ensures the system remains quick and responsive.

Scalability

The module should be able to handle many users at the same time without slowing down.

The AI checks and rules should use resources efficiently, for example by processing inputs in groups.

As the number of users increases, more servers can be added since each check uses very little CPU.

The module should be stateless, meaning it only takes input and gives output, which makes scaling easier.

Maintainability

The code and models should be modular, allowing updates and changes easily.

If new problems arise, the mood model can be retrained without changing the full system.

The setup should allow replacement or updating model files without major modifications to the system.

Security and Privacy

Sensitive data such as mood notes and sensor information must be kept secure.

If any external tools are used for checks, they must follow strict privacy rules.

The current plan avoids sending user data outside the system.

Logs containing flagged issues must also be protected since they may hold personal information.

Only authorized team members should have access to sensitive logs.

Transparency

The system should, if possible, explain its decisions to the user in simple and understandable words.

Even though AI decisions are complex, the user should be informed that these checks are part of the app's rules.

This approach helps build trust and reassures the user that their data is being processed fairly for supportive purposes.

6 DIAGRAMS

6.1 System Diagram

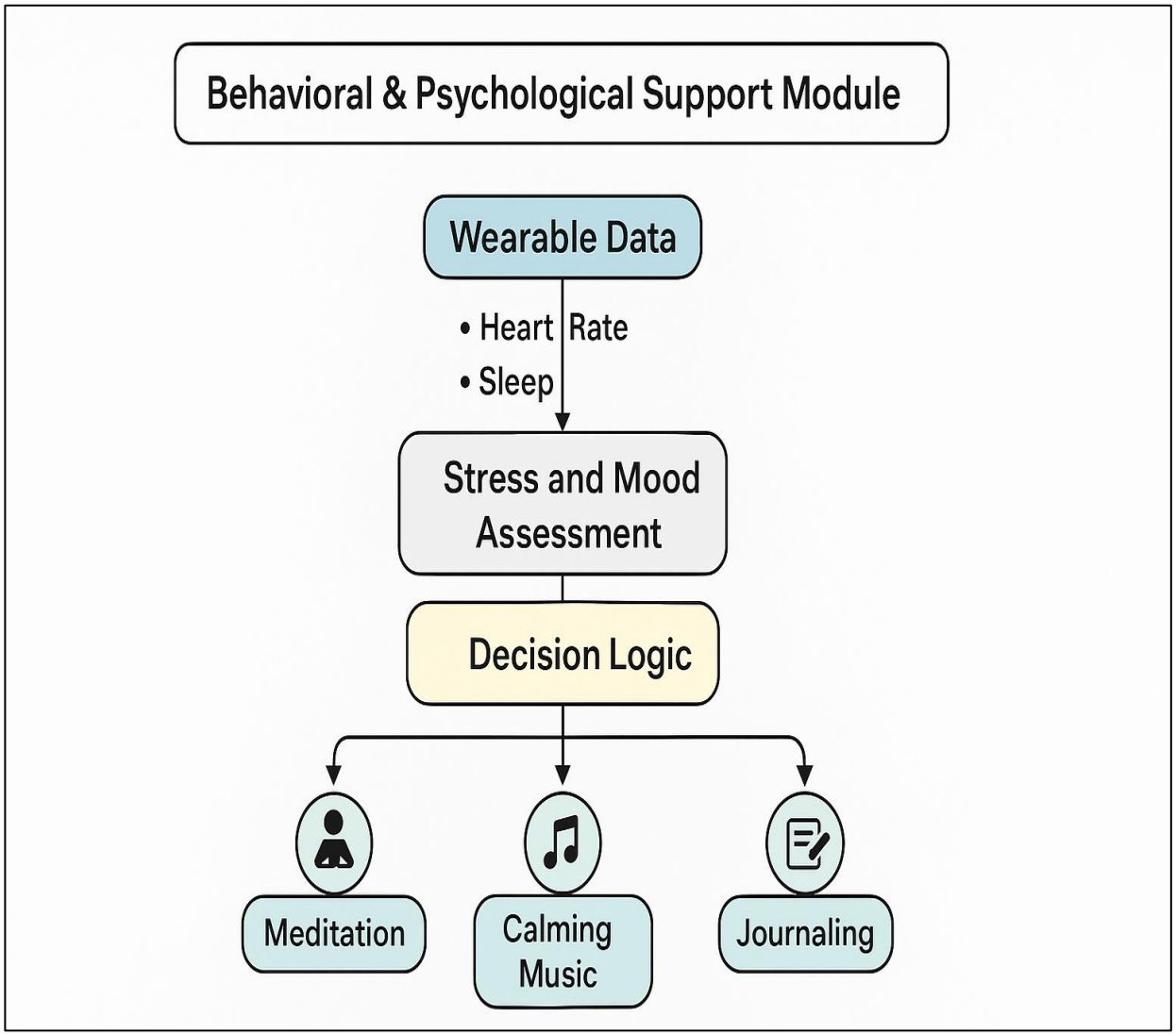

Figure 6.2 System Diagram

6.2 Workflow Diagram

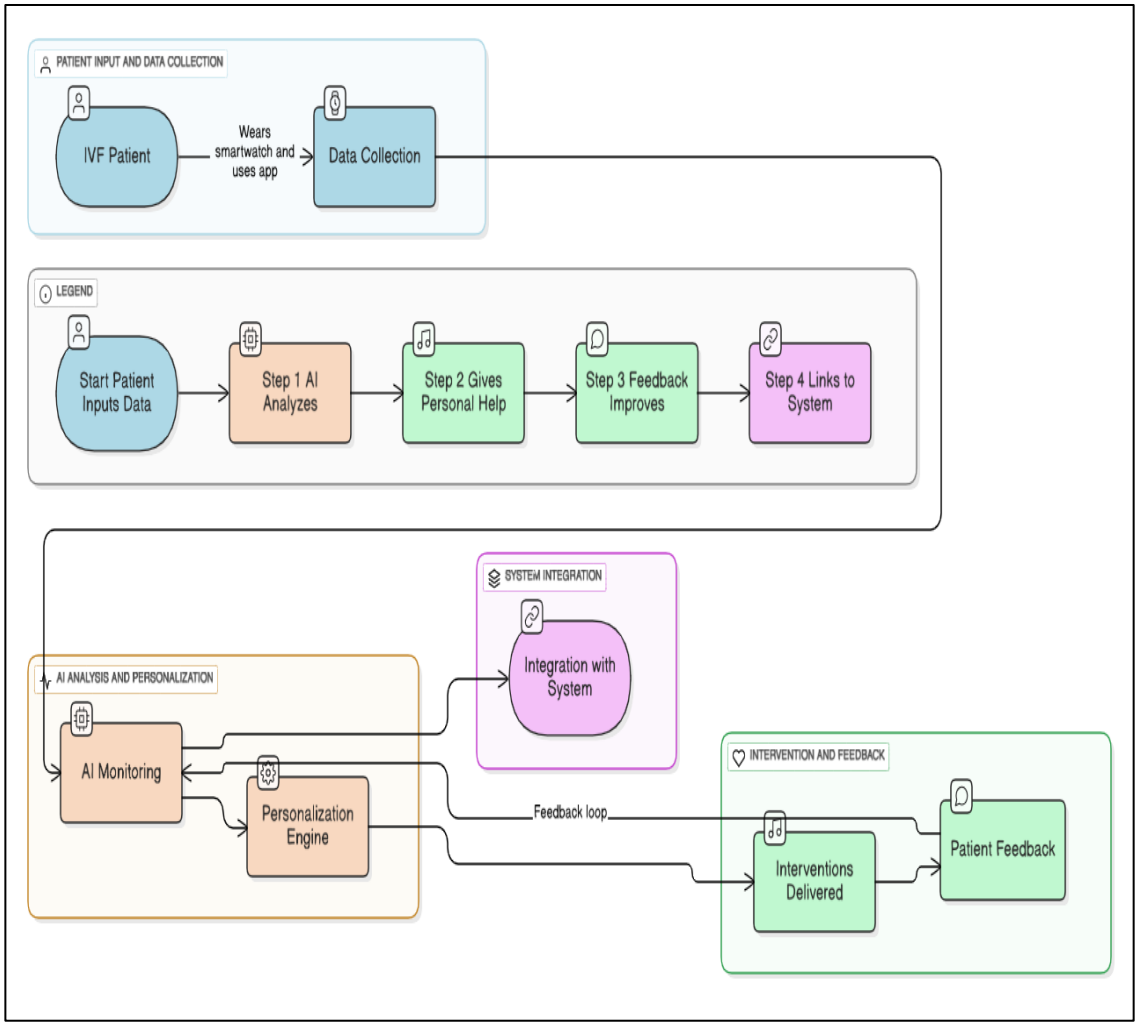

Figure 6.3 Workflow Diagram

7 CONCLUSION

The Behavioral and Psychological Support Module is a vital part of the AI-Powered IVF Lifestyle Support System, bringing a new level of care to IVF patients. By checking body signals and mood, it helps ensure the data is real and accurate, reducing the chance of mistakes that could waste time. This is especially important as stress and sadness during treatment can lower success rates, making mental health support crucial. The module uses sensors, mood tracking, and personal advice to create a trusted system that supports patients every step of the way.

This approach not only improves the reliability of the app but also builds confidence among users. As mental health tools become more common, the need for systems like this will grow. Our module stands out by being the first to focus on clear, trustworthy support in IVF care. It helps patients feel supported and understood, which can make a big difference in their treatment journey.

Looking ahead, this module sets a strong foundation for future improvements in fertility care. By combining technology with personal care, it addresses the emotional challenges of IVF while keeping data accurate. This balance ensures patients receive the best possible support, making the process smoother and more effective. As more people rely on such systems, our module will continue to lead the way in providing reliable and caring assistance, helping to improve outcomes and trust in IVF treatments.

8 BUDGET AND BUDGET JUSTIFICATION

Component	Amount (LKR)	Justification
Data Collection	8,000	Costs for gathering stress, sleep, activity, and mood data from IVF patients.
Travel Expenses	3,000	Travel costs for field visits to collect data or consult with healthcare providers.
Total	11,000	

Table 8.1 Budget Planning for Behavioural and Physiological support module

9 COMMERCIALIZATION

The building of the AI-Powered IVF Lifestyle Support System and its Behavioral and Psychological Support Module gives big business chance. This fixes a growing need for mind check in digital health care.

9.1 Market Opportunity

This tool has big market beyond first IVF use.

Health and Fertility Care: Clinics and apps spend on digital change. Our module helps as key difference for patient apps, stopping low mind from hurting success. This market is big in all countries.

Company and Help Programs: Firms in health or family care may license white-label version. This lets workers get mind help, boosting work and a good company image.

Linked Markets - Mind Tools: The mind check engine can sell as alone service API. Clients like:

Health Apps: For check user mood notes.

Insurance: To fight wrong claims with mind data.

Care Sites: To find and filter low mood in users.

More easy AI tools make need for check fixes, putting our module to fill trust gap in health tech.

9.2 Revenue Model

Many ways to make money are planned:

Service to Clinics: Main way is licensing full system to health places with year fee. Price based on users, reports, and help levels.

Custom B2B Fixes: Firms get branded version. This has one-time set fee plus year license.

API as Service: The mind checks module as alone API. Charge per use (e.g., cost per 1,000) with cuts for big use.

Funds and Joins: Start set and build can come from funds for health clear and new tech. Joins can give first money or help for share or good terms, as check for bigger sales.

This mixed money plan lowers risk by using many clients, letting project grow from IVF app to wider B2B health fix.

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